Sahil B Pillai

Robotics Trainer & STEM Educator | Embedded Systems | Applied AI & IoT

Aroor, Alappuzha, Kerala, India | +91 79942 23905 | sahilpillai117@gmail.com
linkedin.com/in/sahil-b-pillai | github.com/Sahilpillai006

Professional Summary

Robotics Trainer and STEM Educator with hands-on experience teaching electronics, robotics, programming, and drone technology to school and college students. Skilled in lesson planning, curriculum development, and project-based learning using Arduino, Raspberry Pi, and computer vision tools. Strong applied foundation in AI and IoT with the ability to translate complex technical concepts into practical learning outcomes.

Core Skills

Teaching & Education: Lesson Planning, Classroom Management, Curriculum Design, Student Assessment, STEM Education
Programming: Python, Arduino, HTML (Basics)
Robotics & Embedded Systems: Arduino, Raspberry Pi, Sensors & Actuators, IoT Systems
AI & Computer Vision: OpenCV, TensorFlow, Image Processing
Design Tools: Fusion 360, SolidWorks, Tinkercad
Soft Skills: Leadership, Communication, Critical Thinking
Languages: English, Hindi, Malayalam

Professional Experience

PADMANABH Innovations – Mentor Rajasthan, India Mar 2026 – Present

- Mentoring students in robotics, innovation, and project-based STEM learning.
- Guiding learners in problem-solving, prototype development, and technical project execution.
- Supporting hands-on robotics projects involving embedded systems, sensors, and basic automation concepts.

Evolve Robotics – Robotics Trainer & Engineer (Team Lead) Kochi, India Jul 2024 – Mar 2026

- Led and coordinated a team of trainers at Bhavan's Vidya Mandir, Eroor, ensuring smooth execution of robotics programs.
- Acted as the primary point of contact between the school and organization for planning, scheduling, and delivery.
- Delivered hands-on robotics and electronics training to 200+ students across school and college levels.
- Designed structured, project-based robotics curriculum aligned with CBSE and STEM education goals.
- Guided students through Arduino programming, sensors, motors, and introductory AI-based projects.
- Conducted hands-on workshops on 3D modeling (Fusion 360, SolidWorks) and drone technology.
- Supported internal R&D activities including robot prototyping, embedded system testing, and design validation.

Projects

Fish Density Monitoring Drone (Ongoing)
Tools: Pixhawk Flight Controller, Raspberry Pi, Camera Module, Python, OpenCV
Role: Developer
Developing an autonomous drone-based system to estimate fish density in water bodies using aerial imagery and computer vision techniques. The system processes real-time video frames to analyze fish movement patterns and estimate population density for aquaculture monitoring and environmental analysis.
Duration: Oct 2024 – Present
Focus: Robotics Systems, Computer Vision, Environmental Monitoring

Crack Detection using Computer Vision
Tools: Python, OpenCV, Raspberry Pi
Role: Developer
Built a computer vision model to detect structural cracks in surfaces, improving inspection efficiency by 60%.
Focus: Image Processing, Applied AI

Human Detection and Face Recognition System
Tools: Python, TensorFlow, Raspberry Pi
Role: Developer
Designed a vision-based system for detecting humans and performing facial recognition in disaster-response scenarios.
Focus: Computer Vision, Robotics Applications

Vision-based Sign Language Recognition System
Tools: Python, TensorFlow, OpenCV, scikit-learn
Role: Lead Developer
Developed a real-time sign language recognition system using CNNs with over 85% accuracy. Converts hand gestures into text and speech to support accessibility-focused applications.
Focus: AI Education, Computer Vision

INSIGHT – Surveillance Drone
Tools: KK Board, Arduino
Role: Developer
Developed a surveillance drone with live video streaming and a public announcement system for remote monitoring applications.
Focus: Drone Technology, Embedded Systems

Education

Bachelor of Technology in Robotics and Automation Engineering
Toc H Institute of Science and Technology, Kerala, India 2019 – 2023
CGPA: 6.54
Relevant Coursework: Embedded Systems, Artificial Intelligence, Machine Learning, Industrial Automation

Higher Secondary – Bio-Maths
Vaduthala Jama-ath HSS, Kerala State Board 2016 – 2018
Percentage: 75%

Certifications

- Robot Programming – FANUC India
- Industrial Automation (PLC, Hydraulics & Pneumatics)
- Mobile Robotics – Srishti Robotics
- Arduino & IoT Workshop
- Raspberry Pi Technical Training

Leadership & Achievements

- Finalist – Low-Cost AAC Device Challenge (NISH & KTU)
- Runner-up – National Healthcare Hackathon 2021 among 200+ participants
- Treasurer – IEEE SB Toc H, managed budgeting and sponsorships
- Design Lead – IEEE Events, led branding and UI/UX for technical workshops

Keywords

Robotics Trainer, STEM Educator, Robotics Teacher, Embedded Systems, Arduino Programming, Raspberry Pi, Computer Vision, OpenCV, Python Developer, AI Projects, IoT Systems, Robotics Curriculum, Project-Based Learning, 3D Modeling, Fusion 360, Technical Instructor